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4. Fasor dan kompleks

Dik = fasor penjualan

$$Z_1 = (a+2) + (b+1) \cdot i \quad (\text{fasor Produk Unggulan})$$

$$Z_2 = (b-1) + (a+2) \cdot i \quad (\text{fasor Produk Pesaing})$$

a. Konversi z_1 dan z_2 ke $r \cdot \text{cis}(\theta)$, Hitung θ dalam radian dan $^\circ$ (4 angka Setelah koma)

$$\begin{aligned} z_1 &= (3) + (7) \cdot i \\ &= 3 + 7i \\ r &= \sqrt{3^2 + 7^2} = \sqrt{58} \quad \theta = \tan^{-1}\left(\frac{7}{3}\right) \\ &\theta = 1.1659 \text{ rad} = 66.8014^\circ \\ z_1 &= \sqrt{58} \text{ cis}(66.8014^\circ) \end{aligned}$$

$$\begin{aligned} z_2 &= (7) - (3)i \\ r &= \sqrt{7^2 + 3^2} = \sqrt{58} \quad \theta = \tan^{-1}\left(\frac{-3}{7}\right) \\ &\theta = -0.4049 \text{ rad} = -23.1986^\circ \\ z_2 &= \sqrt{58} \text{ cis}(-23.1986^\circ) \end{aligned}$$

b. Hitung Manual $z_1 + z_2$, $z_1 \cdot z_2$, z_1 / z_2 , $|z_1 - z_2|$

$$\begin{aligned} z_1 + z_2 &= (3+7) + (7-3)i = 10 + 4i \\ z_1 \cdot z_2 &= (3+7i) \cdot (7-3i) = 21 - 9i + 49i - 21i^2 = 21 + 40i + 21 = 42 + 40i \\ z_1 / z_2 &= \frac{3+7i}{7-3i} \cdot \frac{7+3i}{7+3i} = \frac{42+9i+49i+21i^2}{49+9} = \frac{21+58i}{58} = 0.3621 + 1i \end{aligned}$$

$$|z_1 - z_2| = -4 + 10i$$

$$= \sqrt{16 + 100} = \sqrt{116}$$

c. De Moivre dan Argabar

$$(z_1)^4 = (\sqrt{58})^4 \cdot \text{cis}(4\theta)$$

$$= 58^2 \cdot \text{cis}(267.2056^\circ)$$

$$= 3364 (\cos 267.2056^\circ + i \sin 267.2056^\circ)$$

$$\cos 267.2056^\circ = -0.0488$$

$$\sin 267.2056^\circ = -0.9988$$

$$= 3364(-0.0488 + 0.9988i)$$

$$= -164 - 3360i$$

Pembuktian

$$1. (3+7i)(3+7i) = -40 + 42i$$

$$2. (-40 + 42i)(3+7i) = (-120 - 280i + 126i + 294i^2) = -120 - 154i - 294 = -414 - 154i$$

$$4. (-414 - 154i)(3+7i) = (-1242 - 2898i - 462i - 1078i^2) = -1242 - 3360i + 1078 = -164 - 3360i$$

Hasil Perbandingan Sama $\approx$ menunjukkan $= -164 - 3360i$

(KIKI)

D. Ahar Pangkat 3

$$D: u = w = 8 \text{ cis}(60^\circ)$$

$$\text{Ahar} = z_k = 2 \text{ cis} \left(\frac{60 + 360k}{3} \right)$$

$$k=0: z_0 = 2 \text{ cis} \left(\frac{60 + 360 \cdot 0}{3} \right) = 2 \text{ cis}(20^\circ)$$

$$k=1: z_1 = 2 \text{ cis} \left(\frac{60 + 360 \cdot 1}{3} \right)$$

$$= 2 \text{ cis} \left(\frac{420}{3} \right) = 2 \text{ cis}(140^\circ)$$

$$k=2: z_2 = 2 \text{ cis} \left(\frac{60 + 360 \cdot 2}{3} \right)$$

$$= 2 \text{ cis}(260^\circ)$$

Ubah ke atbi

$$z_k = \cos \theta + i \sin \theta$$

$$= 2 (\cos 20^\circ + i \sin 20^\circ) \approx 1.879 + 0.684i$$

$$z_2 = \cos \theta + i \sin \theta$$

$$= 2 (\cos 140^\circ + i \sin 140^\circ) \approx -1.532 + (2.088)i$$

$$z_3 = \cos \theta + i \sin \theta$$

$$= 2 (\cos 260^\circ + i \sin 260^\circ) \approx -0.347 - 1.970i$$

E. Interpretasi

= Arg (z_1/z_2) menunjukkan perbedaan fase antar produk dalam penjualan Mingguan.

Yaitu jika nilainya besar, artinya ada pergeseran waktu penjualan antar

Produk, contoh: Satu produk laku lebih dulu dibanding yang lain.

Hal ini penting sebagai strategi pemasaran seperti penjadwalan promosi.

Fourier transform penting karena dapat mengubah data deret waktu

menjadi domain frekuensi sehingga pola musiman